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SYSTEMATIC REVIEW

Patient Preferences Regarding Menopausal Symptoms and Treatments: A Systematic Review of Quantitative Stated-Preference Studies

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Abstract

Objective Ensuring therapies align with women's expectations can improve adherence and is crucial for the effectiveness and safety of menopausal treatments, especially in light of the concerns raised after the Women's Health Initiative study. This systematic review aims to provide an overview of studies reporting data on women's preferences for menopausal symptoms and treatments to better guide research and development of therapies.

Design A systematic literature review was conducted on Medline (via Ovid) and Embase. This review used a structured search strategy to identify all quantitative stated-preference evidence on menopausal symptoms or treatments, including conjoint analysis, discrete choice experiment, best-worst scaling, or quantitative preference survey, published before April 2025. A manual search complemented the process. The quality of the included conjoint analyses was assessed using the Discrete Choice Experiments Reporting Checklist (DIRECT), and quantitative surveys were evaluated using the Consensus-Based Checklist for Reporting of Survey Studies (CROSS) checklist. The whole conduct of the systematic review was performed in adherence to the Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) 2020 statement (PROSPERO registration no. CRD42024614218).

Results In total, 606 references were screened after removing duplicates, and 7 studies were included (4 conjoint analyses and 3 quantitative surveys). Five studies were conducted in the USA, one in Sweden, and another was conducted across five European countries and the USA. Two main topics were covered: (1) preferences regarding menopausal symptoms and treatments and (2) preferences toward local estrogen therapies. First, regarding preferences for menopausal treatment, the most important attribute categories were efficacy and long-term safety. Limited data were provided on other treatment characteristics. Studies reported that women are dissatisfied with their current therapeutic options. Second, two studies focused on local estrogen therapies and revealed a preference for vaginal tablets with applicators over other treatment modalities.

Conclusions Research on menopausal treatment preferences is currently limited. Diverse study designs hampered the comparison and synthesis of the results. There is a need to conduct further conjoint analyses, such as discrete choice experiments (DCE), to better understand women's preferences and improve patient management and treatment options.

Key Points for Decision Makers

Regarding menopausal symptoms and treatments preferences, significant methodological heterogeneity exists within conjoint analyses and within quantitative surveys, making comparisons challenging.

Among menopausal women, the most important attribute categories and items were efficacy and long-term safety.

Further DCEs with comprehensive attribute sets to cover the full range of menopause care are needed in Europe, Asia, Africa, Oceania and Latin America.

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1 Introduction

Menopause marks the permanent cessation of ovarian function due to the depletion of ovarian follicles, typically occurring between the ages of 42 and 58 years [1]. As life expectancy increases, women are now spending a growing portion of their lives in a postmenopausal state. Over the past decade, the share of women aged 50 and over rose from 22 to 26% of all women and girls globally [2].

Menopausal symptoms can begin in the late reproductive stage, when fertility starts to decline. Some women start to experience various symptoms, including vasomotor symptoms (VMS; e.g., hot flushes and night sweats), breast tenderness, insomnia, migraines, and premenstrual dysphoria during this period [1]. In the later postmenopausal years, genitourinary symptoms become more prominent, such as vaginal dryness and urinary symptoms, while signs of somatic aging become increasingly noticeable [1, 3]. Over the long term, estrogen deficiency also contributes to bone demineralization, neurocognitive changes, and increased cardiovascular risk [4]. The experience of menopause varies greatly from one woman to another. Moreover, menopausal symptoms vary across the transition, even within the same individual. This variability highlights the complex interplay of biological, psychological, and sociocultural factors. Among all symptoms, VMS are the most common and are associated with decreased sleep quality, negative mood, and overall quality of life. The Study of Women's Health Across the Nation (SWAN), a large, multiethnic longitudinal cohort, found that up to 80% of women experience VMS during the menopausal transition, and that these symptoms may persist for over 8 years in some individuals [5].

In 2002, the Women's Health Initiative (WHI) study revealed an increased risk of cardiovascular events and breast cancer with menopausal hormone therapy (MHT). These results led to a dramatic reduction in their use [6], and many women were left without informed care [7]. The initial conclusions of the WHI, which were based on inappropriate interpretations, led to an important societal impact by distorting the true benefit–risk balance of MHT. Recently, the US Food and Drug Administration removed the black-box warnings for some MHT, an important step toward correcting misinformation from the WHI and promoting a more evidence-based understanding of its risks and benefits [8]. In addition, new hormonal and nonhormonal therapies have emerged over the past two decades. Ongoing research is exploring safer MHT options, such as body-identical estrogens (e.g., estetrol), as well as novel nonhormonal treatments, including neurokinin B receptor antagonists [9, 10]. While MHT remains the gold standard for relieving menopausal symptoms and preventing deleterious conditions associated with menopause, such as osteoporosis and cardiovascular disease, hormonal and nonhormonal treatment strategies should be individualized and adapted according to women's medical history as well as their personal preferences [11]. Taking patient preferences into account at every stage of decision-making can lead to more acceptable and effective healthcare strategies, eventually improving treatment initiation, acceptance, satisfaction, compliance, and persistence [12–14].

Methods for evaluating stated preferences, such as conjoint analysis, are becoming more prevalent in healthcare. Such studies help ensure that a treatment choice meets women's expectations, thereby improving compliance and, consequently, the efficacy and safety of menopausal treatments [15]. This systematic literature review aims to synthesize findings from studies (i.e., conjoint analyses and quantitative surveys) on women's preferences regarding menopausal symptoms and treatments to better understand these preferences and guide research and interventions in the field.

2 Materials and Methods

The proposed systematic review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) 2020 statement. The completed PRISMA checklist is available in the Electronic Supplementary Material. A protocol has been developed and published in PROSPERO (CRD42024614218).

2.1 Literature Search and Study Selection

A search strategy, developed in collaboration with an experienced researcher (C.B.), was applied in October 2024 and updated in April 2025 on two bibliographic databases, Medline (via Ovid) and Embase, to identify any quantitative stated preferences studies (conjoint analyses and quantitative surveys) providing women's preferences regarding menopause symptoms and treatment characteristics. The two search strategies are available in the Electronic Supplementary Material. Besides electronic bibliographic searches, experts in the fields were contacted, and the bibliographies of included studies were investigated to identify any potential additional studies. The search results from the electronic sources and hand searching were imported into Covidence software for data management. Covidence is a web-based collaboration software platform that streamlines the production of systematic and other literature reviews. During the initial screening phase, two reviewers (C.B. and L.R.) independently assessed the title and abstract of each reference to eliminate irrelevant articles. Rigorous inclusion criteria were applied, as presented in Table 1. In the subsequent step, the two reviewers individually (C.B. and L.R.) examined the

full text of each article that was not excluded in the initial stage, selecting studies that fulfilled the inclusion criteria. Any discrepancies in article selection were resolved through discussion and consensus.

2.2 Data Extraction

All included studies were summarized using a data extraction form (Excel sheet) developed on the basis of previous research and adapted for this systematic review. The extraction form was tested on two studies before the extraction began. Two reviewers (L.R. and E.B.) independently extracted the data. Any discrepancies were resolved through discussion. General study information, population characteristics, methodological details, and main results were extracted (Table 2).

2.3 Quality Assessment

Two reviewers (L.R. and E.B.) independently assessed the quality of included studies using the Discrete Choice Experiments Reporting Checklist (DIRECT) [16] for conjoint analyses and the Consensus-Based Checklist for Reporting of Survey Studies (CROSS) [17] for quantitative surveys. Any discrepancies were resolved through discussion.

The DIRECT checklist includes 26 items that consider multiple dimensions of quality, such as the appropriateness of the research question and the clarity and robustness of the study design. It evaluates items related to the purpose and rationale, attributes and levels used in the experimental design, the design and piloting of the survey, sample and data collection, the econometric analysis, and the reporting of the results. Each item was assessed and rated from 0 points (item not satisfied) to 2 points (item fully satisfied). Therefore, studies could reach a maximum score of 52 points.

The CROSS checklist can be applied to both web- and nonweb-based surveys. The CROSS checklist includes 19 main items and 40 individual criteria that encompass study main characteristics, design, sampling, data collection, survey administration, statistical analysis, and reporting of results. Therefore, studies could reach a maximum score of 72 points (this maximum score applies when the study has no multivariate analysis, no longitudinal component, and no sensitivity analyses).

2.4 Data Analysis

The conjoint analysis attributes and quantitative survey items were analyzed for frequency and similarity across all the included studies. Because methodologies differ significantly from one article to another, direct comparisons between conjoint analyses and quantitative surveys were not possible. For the purposes of comparison and synthesis, different categories have been established for conjoint analyses and quantitative surveys. First, conjoint analyses were analyzed and categories of attributes were determined. A conditional relative importance analysis of the reported attributes was not performed because the necessary information was reported in only one study. Second, to enable comparison between conjoint analysis attribute categories and quantitative survey items, quantitative survey items were classified into categories aligned with those used in the conjoint analysis. Items from quantitative surveys were first grouped into categories, then into subcategories. Alluvial diagrams were developed to allow a visual representation of the items (© 2024 Flourish). To integrate the conjoint analysis and quantitative survey results, radar charts were constructed (© 2024 Flourish). An overall radar chart was generated that integrated all conjoint analysis and quantitative survey results, using weights that reflected the relative contributions of each source. Studies that investigated the preferences of local estrogen therapies (LET), conjoint analyses and quantitative surveys, were analyzed separately.

Table 1 Inclusion and exclusion criteria

	Inclusion
Population	Women of any demographic (i.e., age, ethnicity) in perimenopause, menopause, or post menopause
Intervention and exposure	Menopausal symptoms Any menopausal therapy (hormonal, nonhormonal)
Outcome	Menopausal symptoms and treatments preferences Value of menopausal symptom relief Quantitative treatment preferences
Study design	Quantitative preference study (e.g., discrete choice experiment, quantitative survey)
Language	English* French*

*https://PubMed.ncbi.nlm.nih.gov/22559755/; https://www.ScienceDirect.com/science/article/pii/S0895435621001347

Table 2 Study characteristics

References	Funding	Study design	Country	Participants	Investigated preference	Patients' mean age
Preferences regarding VMS treatments						
Johnson et al. [19, 20]	Wyeth Pharmaceuticals	Conjoint analysis (rating scale conjoint analysis)	USA	Women, 46–60 years, resident in USA	Preferences regarding VMS treatments	52 years
Craig et al. [24]	Dr. Craig's support account at Moffitt Cancer Center	Conjoint analysis (mixed approach*)	USA	Women 40–69 years	Value of menopausal symptoms relief	NI Two age groups 40–54 years 55–69 years
Stute et al. [22]	Astellas Pharma, Inc.	Quantitative survey	UK, France, Germany, Italy, Spain, USA	Patients: women, 40–65 years with clinically confirmed VMS associated with menopause Physicians: responsible for treatment decisions and to see at least three patients with VMS per month (primary care providers and gynecologists)	Impact of menopausal symptoms, treatment patterns, and perceptions	Patients: 53.9 years Physicians: NI
Shiozawa et al. [25]	Astellas Pharma, Inc.	Conjoint analysis (DCE)	USA	Women, 40–65 years; perimenopausal with a uterus and at least one ovary or naturally postmenopausal with a uterus and at least one ovary, or surgically postmenopausal with a hysterectomy with bilateral oophorectomy, and experienced at least 14 hot flashes and/or night sweats during at least 1 week in the month before they were due to complete the survey	Preferences regarding VMS treatments	53.7 years
DePree et al. [26]	Astellas Pharma, Inc.	Quantitative survey	USA	Patients: women, 40–65 years, perimenopausal or postmenopausal, who experienced moderate-to-severe VMS in the past 12 months, received at least one treatment for VMS in the past 3 months Physicians: obstetricians and gynecologists, primary care providers licensed and actively practiced medicine in the USA, had treated at least 15 women with VMS in the past 3 months, had prescribed treatment for women with VMS in the past 3 months	Treatment satisfaction and expectations, unmet needs for VMS treatments	Patients: 53,1 years Physicians: 52.9 years
Preferences regarding LET						
Mattsson et al. [21]	Novo Nordisk	Conjoint analysis (DCE)	Sweden	Women > 50 years, resident in Sweden, postmenopausal; experience of menopausal change in and around the vagina; use of LET	Preferences toward attributes of LET	61.1 years

Table 2 (continued)

References	Funding	Study design	Country	Participants	Investigated preference	Patients' mean age
Minkin et al. [23]	Novo Nordisk	Quantitative survey	USA	Women > 18 years, postmenopausal, or bilateral oophorectomy, currently receiving LET in the form of vaginal tablets, to have a prior experience using other formulations of LET	Women's satisfaction with LET; reasons for switching to vaginal tablets	NI 52% between 35 and 57 years and 48% ≥ 58 years

DCE discrete choice experiment, *LET* local estrogen therapy, *NI* no information, *VMS* vasomotor symptoms

*Craig et al. used a mixed approach: Respondents completed a checklist that assesses 30 domains of menopausal symptoms. Each response on this checklist describes the health outcome of the paired comparisons. Each respondent had to choose between one of the symptoms and a lifespan loss with no health problems (i.e., lifespan pairs) for eight pairs. Then, respondents chose between two symptoms for 22 pairs (i.e., health pairs)

3 Results

3.1 Study Characteristics

Systematic database searches yielded 750 studies. As shown in Fig. 1, 144 were duplicates and consequently removed from the screening, 606 articles were screened on the basis of the title and the abstract, and 16 articles were screened on the basis of their full texts. In total, six articles, four conjoint analyses, and two quantitative surveys met our inclusion criteria. The reasons for the exclusion of the ten other articles are shown in Fig. 1. Although it was eligible according to the inclusion criteria, one study was excluded owing to a minor protocol deviation related to significant differences in the population (Cambodian women in community settings), to maintain comparability between studies and to ensure the validity of the resulting comparisons [18]. In addition, the manual search identified two studies (one conjoint analysis and one quantitative survey).

In total, eight studies were therefore included in the systematic review: five conjoint analyses and three quantitative surveys. Two of these conjoint analyses (i.e., Johnson et al. [19, 20]) were published by the same author, included the same population, and used the same questionnaire. Consequently, these two articles were treated as a single study.

3.2 Quality Assessment Based on DIRECT and CROSS Checklists

On average, the included conjoint analyses scored 42 of 52 possible points in the DIRECT checklist (Table 3). Across all four articles, the highest scores were obtained in *purpose and rationale* (mean 4/4) and *attributes and levels rationale* (mean 4/4). The lowest-scoring category was *experimental design* (mean 7.5/12). The lowest-ranked items were (1) indicate how the experimental design was obtained and (2) report the model performance and goodness of fit (if comparing models). Overall, Johnson et al. achieved the highest total score (50/52) [19, 20], and Mattsson et al. [21] the lowest (31/52). The completed DIRECT checklist is provided in the Electronic Supplementary Material.

Regarding the quality assessment of quantitative surveys, the included studies averaged 52/72 points on the basis of the CROSS checklist (Table 4). No study indicated whether methods such as item weighting or propensity scores were used to adjust for sample non-representativeness. The lowest-scoring items were (1) stating how nonresponse error was addressed (0.67/2) and (2) providing information on the data-entry process (0.33/2). Overall, Stute et al. achieved the highest total score (57/72) [22] and Minkin et al. [23] the lowest (46/72). The completed CROSS checklist is provided in the Electronic Supplementary Material.

3.3 Study Population

The seven included studies looked at perimenopausal to postmenopausal women, with similar target age ranges (ranging from 40 to 69 years), except for one study, which included younger women (over the age of 18) [23]. Five studies focused on women in the USA, one study examined residents of Sweden, and another was conducted across five European countries and the USA (Table 2).

3.4 Systematic Menopausal Hormonal Therapy and Menopausal Symptoms

3.4.1 Study Method

A total of three conjoint analyses and two quantitative surveys were identified. Among the conjoint analyses, one used discrete choice experiments (DCE), another employed rating-scale conjoint analysis, and a third utilized a mixed method combining adaptive and partial profile paired approach comparisons (Table 3). The process of selecting attributes and levels was outlined in all three studies. All attributes were chosen on the basis of a literature review or prior research [19, 20, 24, 25]. In addition, attribute selection was further informed by a telephone interview in one study [25] and by consultation with experts in another [19, 20]. Two studies carried out pretesting, which took the form of interviews [19, 20] or virtual meetings [25]. Conjoint analyses contained between three and seven attributes and between three and five levels. The total number of choice sets across the studies ranged from 27 to 42, with between 9 and 30 sets per participant.

Regarding the two quantitative surveys, the instruments used in Stute et al. included the Menopause Specific Quality Of Life Questionnaire (MENQOL) to assess participants' quality of life, and the Work Productivity and Activity Instrument (WPAI), a questionnaire adapted for hot flashes or night sweats, to evaluate the impact of VMS on activity and productivity [22]. In addition, DePree *et al.* used the Menopause Symptom Treatment Satisfaction Questionnaire (MS-TSQ), which assessed treatment satisfaction among women and physicians [26].

3.4.2 Conjoint Analyses

Figure 2 shows the distribution of the 17 attributes reported in the 3 conjoint analyses assessing menopausal symptoms and treatment preferences [19, 20, 24, 25]. Three attribute categories were identified: efficacy, safety, and cost attributes. The majority of attributes were classified under efficacy attributes (58.82%), followed by safety attributes (35.29%) and cost attributes (5.88%). Attributes under the efficacy category included frequency, severity, duration of symptoms, hot flushes and/or night sweats, and impact on quality of life (QoL) and activities of daily living (ADL). Safety attributes focused on long-term side effects such as cardiovascular

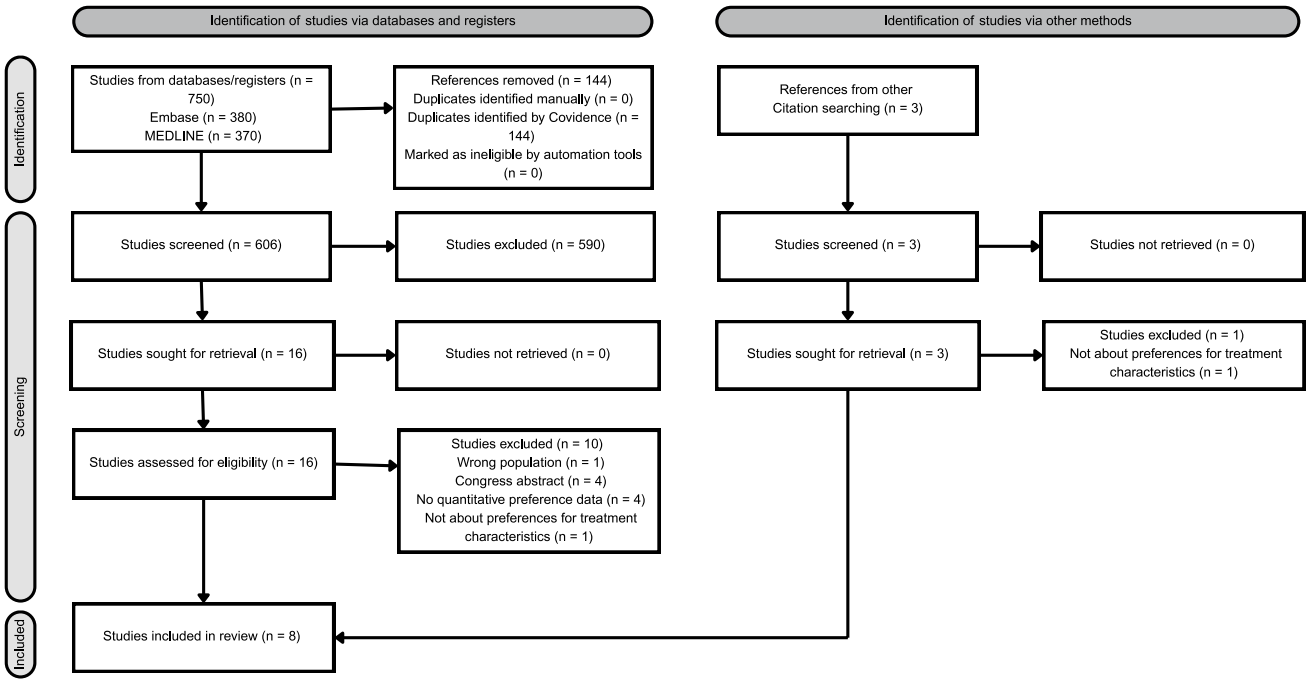


Fig. 1 PRISMA flowchart. A total of eight studies were identified through Medline and Embase and manual searches: six conjoint analyses and two quantitative surveys. Two of these conjoint analyses (e.g., Johnson et al. [19, 20]) were considered and analyzed as a single study

Table 3 Study methods for conjoint analyses

References	Study design	Selection process of attributes and levels	No. of attributes, levels, alternatives, blocks, choice sets	Experimental design type and approach	Administration method	Methods to test comprehension	Analysis methods (model)	Preference heterogeneity	Analysis of model results	DIRECT checklist score
Preferences regarding VMS treatments										
Johnson et al. [19] Johnson et al. [20]	Conjoint analysis (rating scale conjoint analysis)	Review of literature and consultations with experts	7 attributes Maximum of 4 levels 27 choice sets 2 alternatives 3 blocks	Fractional design Choice task unlabeled Near-optimal experimental design	Online	Yes	Random utility model	Yes	MARs HYEs	50/52
Craig et al. [24]	Conjoint analysis (mixed approach*)	Review of existing instruments	3 attributes 5 levels 30 choice sets 2 alternatives Blocks: 30 checkboxes domain-routed[1]	Fractional design Choice task unlabeled	Online	No	Generalized linear model	No	QALYs	40/52
Shiozawa et al. [25]	Conjoint analysis (DCE)	Review of literature and interviews	7 attributes Maximum of 3 levels 2 alternatives 42 choice sets 3 blocks	Fractional design Forced-choice design Choice task unlabeled	Online	Yes	Random parameter logit regression models	No	WTP	48/52
Preferences regarding LET										
Mattsson et al. [21]	Conjoint analysis (DCE)	Attributes defined independently, not matched to specific products	4 attributes Maximum of 3 levels 2 alternatives Choice sets: NI	Choice task labeled	Online	Yes	Conditional logit model	Yes	WTP	31/52

MAR maximum acceptable risk, *WTP* willingness to pay, *QALY* quality-adjusted life years, *HYEs* healthy-year equivalents, *NI* no information

*Craig et al. used a mixed approach: Respondents completed a checklist that assesses 30 domains of menopausal symptoms. Each response on this checklist describes the health outcome of the paired comparisons. Each respondent had to choose between one of the symptoms and a lifespan loss with no health problems (i.e., lifespan pairs) for eight pairs. Then, respondents chose between two symptoms for 22 pairs (i.e., health pairs)

[1]Checkboxes domain-routed, screening checkboxes for each symptom domain: if checked, follow-up items assess frequency, severity, and interference; if unchecked, the domain is skipped

events, cancer, fractures, as well as short-term side effects. The cost attribute was reported only in Shiozawa et al. [25] to perform a willingness-to-pay analysis.

Figure 3a shows the significance of these attributes. The efficacy (including the frequency and severity of VMS and their impact on QoL and ADL) and the long-term safety category are identified as the most important for women.

Regarding efficacy, Shiozawa et al. reported improvement of sleep (e.g., impact on QoL and ADL), reduction in the number of hot flushes and/or night sweats, and reduction in severity of hot flushes and/or night sweats, as the most important attributes (Fig. 4) [25]. Craig et al., which assessed only efficacy attributes, found that VMS frequency was the most important attribute among women [24]. Conversely, Johnson et al. reported that efficacy attributes were less important than safety (Fig. 4). However, the attributes analysis applied by Johnson et al. also revealed that women prefer to trade years of life rather than facing intense menopausal symptoms and trade longer durations of milder symptoms for shorter durations of severe symptoms [20].

Table 4 Study methods for quantitative surveys

Reference	Instruments, questionnaires used	Number of questionnaires	Administration method	Survey content overview	CROSS checklist score
Preferences regarding VMS treatments					
Stute et al. [22]	Patient survey: MENQOL, WPAI	2 One questionnaire for patients One questionnaire for physicians	Paper-based survey for patients Online survey for physicians	Impact of VMS on work/study, sleep, mood, overall QoL Physicians' and patients' perceptions of treatments Physicians' and patients' level of satisfaction with symptom control Patients completed MENQOL and WPAI questionnaires	57/72
DePree et al. [26]	MS-TQ	2 One questionnaire for patients One questionnaire for physicians	Online survey for patients and physicians	Qualitative interviews Quantitative surveys: Physicians' and patients' level of satisfaction with treatment using the MS-TQ questionnaire Physicians' and patients' unmet needs Physicians' and patients' expectations for new VMS treatments	54/72
Preferences regarding LET					
Minkin et al. [23]	NI	1	Online	Patients had to choose the most bothersome symptoms of vaginal atrophy from a list Reasons for missing doses/delayed filling Reasons for switching from cream/ring to vaginal tablets Patients rated how user-friendly tablets/cream/ring were on a 7-point Likert scale Patients selected positive attributes of vaginal tablets Patients were also asked if they were aware that regulatory agencies and medical societies have recommended using the lowest effective dose of estrogen to alleviate symptoms of vaginal atrophy	46/72

MENQOL Menopause-Specific Quality Of Life Questionnaire, *MS-TQ* Menopause Symptom Treatment Satisfaction Questionnaire, *WPAI* Work Productivity and Activity Instrument-specific health problem, *NI* no information

Regarding safety, on the one hand, Shiozawa et al. reported that the risk of breast cancer and cardiovascular events, as well as short-term side effects, were less important among women than sleep improvement and VMS frequency and severity (e.g., efficacy) (Fig. 4) [25]. On the other hand, according to Johnson et al., safety attributes, particularly heart attack risk, were most important for women rather than the frequency of night sweats (e.g., efficacy). However, the authors also reported that women who are well informed about possible risks of MHT, many of whom have experienced VMS, appeared willing to accept explicit tradeoffs between risks (e.g., heart attack and breast cancer) within the ranges reported in the WHI and clinically realistic health state improvements [19].

3.4.3 Quantitative Surveys

A total of two quantitative surveys were included [22, 26]. DePree et al. presented a report on satisfaction and unmet needs with current treatments, as well as expectations for future treatments (Table 4) [26]. Stute et al. reported the impact of VMS in terms of frequency and severity for QoL, sleep, mood, and work/study, and also applied the MENQOL and WPAI tools [22]. Stute et al. also assessed perceptions of treatments and levels of satisfaction with symptom control (Table 4) [22].

The items "expectations" and "unmet needs" shown by DePree et al. were categorized for comparison purposes with the conjoint analysis attributes (Fig. 5) [26]. Four categories were defined: efficacy, safety, cost, and administration. The methodology used by Stute et al. was too different and therefore is analyzed separately [22].

The efficacy criteria (frequency and severity of VMS and impact on QoL and ADL) were the most frequently reported among unmet needs with current treatments and expectations for new treatments (Figs. 3b, 5). DePree et al. reported that the lack of effectiveness was the major unmet need in actual treatments among women. The most reported expectations for new treatments were improvement in sleep and reduction of severity and frequency of hot flushes and/or night sweats. Women reported safety items as the second most unmet need and the third highest expectation for future treatments. The third and fourth unmet needs of current treatment reported by women were long-term safety

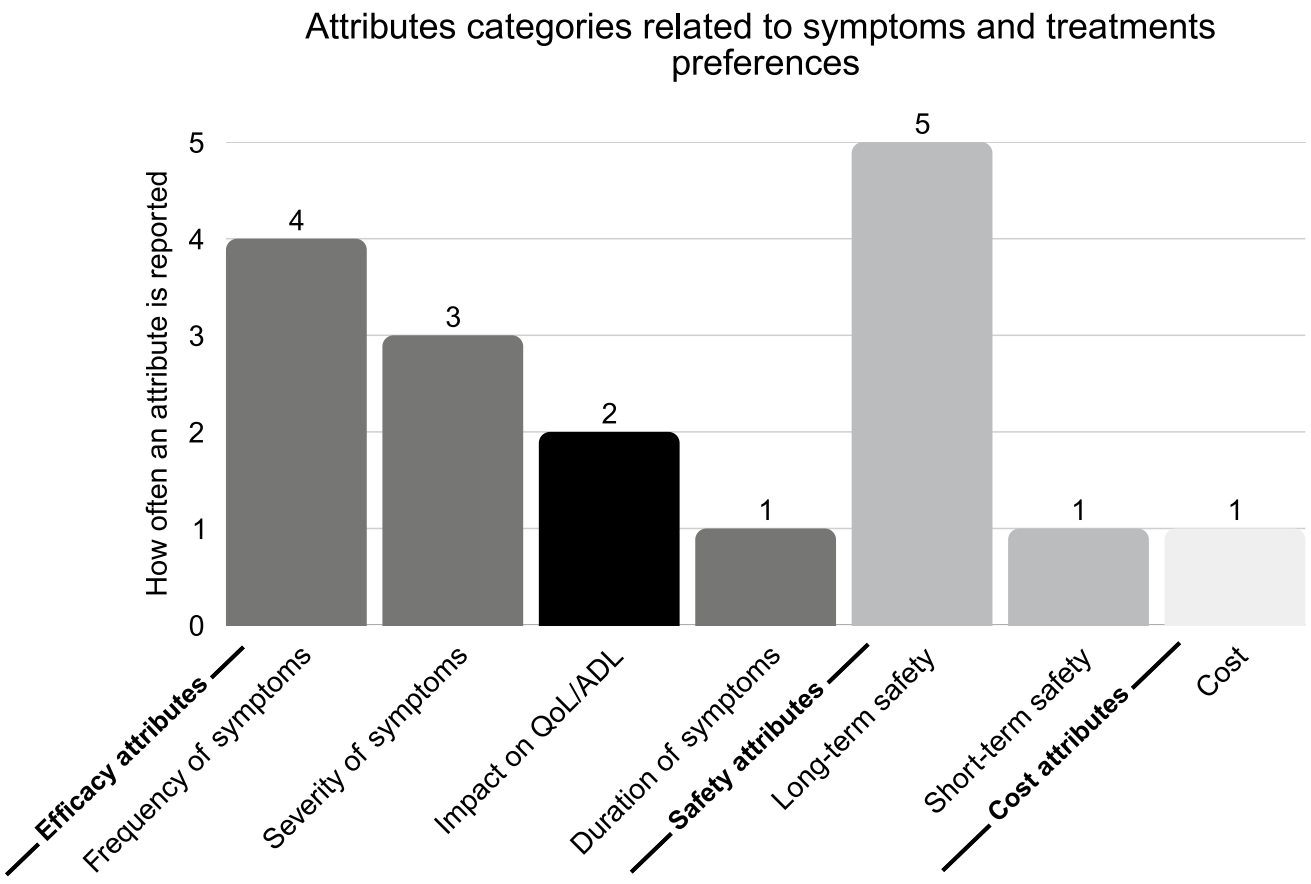


Fig. 2 Attributes categories from conjoint analyses. Symptom and treatment preferences: a total of 17 attributes were extracted from 3 conjoint analyses. Three attribute categories: efficacy, safety, and cost attributes

concerns and lack of information about long-term safety, respectively [26].

Stute et al. also reported that VMS in terms of frequency and severity had the greatest impact on sleep (35.8%) but also influenced mood (31.6%), QoL (23.6%), and work/study (15.4%) [22].

These two quantitative surveys also revealed low satisfaction with current treatments. In both Stute et al. and DePree et al., around one-third of women reported being dissatisfied with their treatment [22, 26]. DePree et al. also showed a similar level of satisfaction among different treatment classes (hormonal and nonhormonal treatments and over-the-counter medications) [26].

3.5 Local Estrogen Therapy and Genitourinary Symptoms

3.5.1 Study Method

One conjoint analysis using the DCE method and one quantitative survey assessed preferences regarding LETs. The included DCE featured four attributes with a maximum of three levels; however, the process for selecting attributes and levels was not clearly reported [21] (Table 3).

3.5.2 Conjoint Analysis

One DCE assessed preferences toward attributes of LET [21]. The most common reason given by women for using LET was vaginal dryness. Four attributes were investigated: mode of administration, smudges/leakage, time of appliance, and cost per month to assess the willingness to pay of women. As shown by their willingness to pay, women preferred small vaginal tablets with a disposable applicator to vagitories and vaginal creams administered with a dosing syringe. Vagitories were preferred to creams. Preferences were based on mode of administration and smudges/leakage, while time of appliance had no significant impact on preferences.

3.5.3 Quantitative Survey

Minkin et al. investigated symptoms of vaginal atrophy, reasons for missing doses, and switching from cream or ring to vaginal tablets [23]. Patients also selected positive attributes of vaginal tablets (e.g., effectiveness in relieving symptoms, convenience of use, neat, clean, and not messy to apply), which were categorized into six categories for comparison purposes: administration, safety, efficacy, smudges/leakage, cost, and time of application.

The most bothersome symptom of vaginal atrophy reported was vaginal dryness. Figure 6 shows that the mode of administration was the most frequently cited positive aspect of vaginal tablets, followed by safety.

The most important reasons reported for switching from vaginal cream to vaginal tablets were the messiness of filling and inserting the applicator, cream leakage, and the unpleasant texture of the cream. Vaginal tablets were rated the most user friendly, followed by vaginal ring and cream.

4 Discussion

This systematic review aimed to investigate women's preferences toward menopausal symptoms and treatments and to identify potential research gaps in the field. The seven

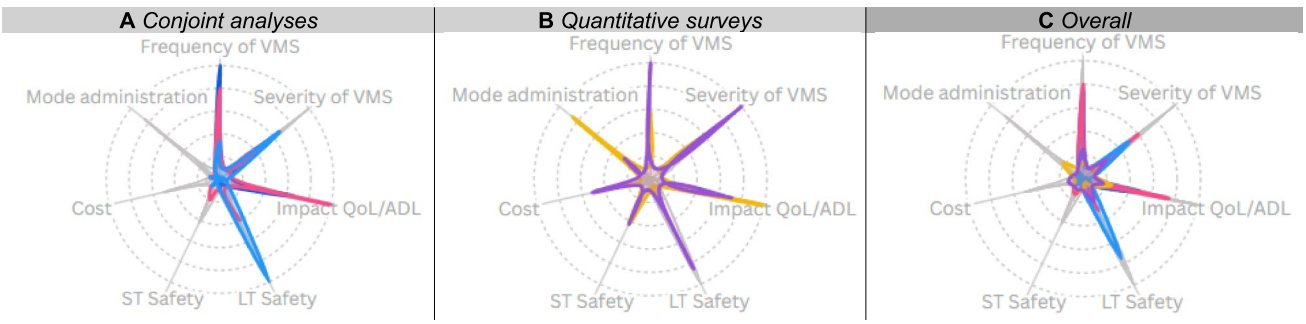


Fig. 3 Overall preference profiles regarding symptom and treatment preferences. An internal scoring scale was established for each article, with a maximum of 7 points depending on the number of attribute categories included (see attribute definitions). For example, Craig et al., addressing only efficacy attributes, had a 3-point scale; Shiozawa et al., covering five categories, had a 5-point scale. The most important attribute received the highest score, the second received one point less, and so on. Scores were then normalized as percentages. This representation is purely visual and does not display preference weights. Instead, it summarizes which attributes were identified as most important in each study, on the basis of an internal ranking system. **A** Radar chart showing the preference results from the conjoint analyses. **B** Radar chart showing the preference results from the quantitative surveys. **C** The overall radar chart combines all conjoint analysis and quantitative survey results, weighted according to their relative contribution (e.g., three conjoint analysis sources versus one quantitative survey source)

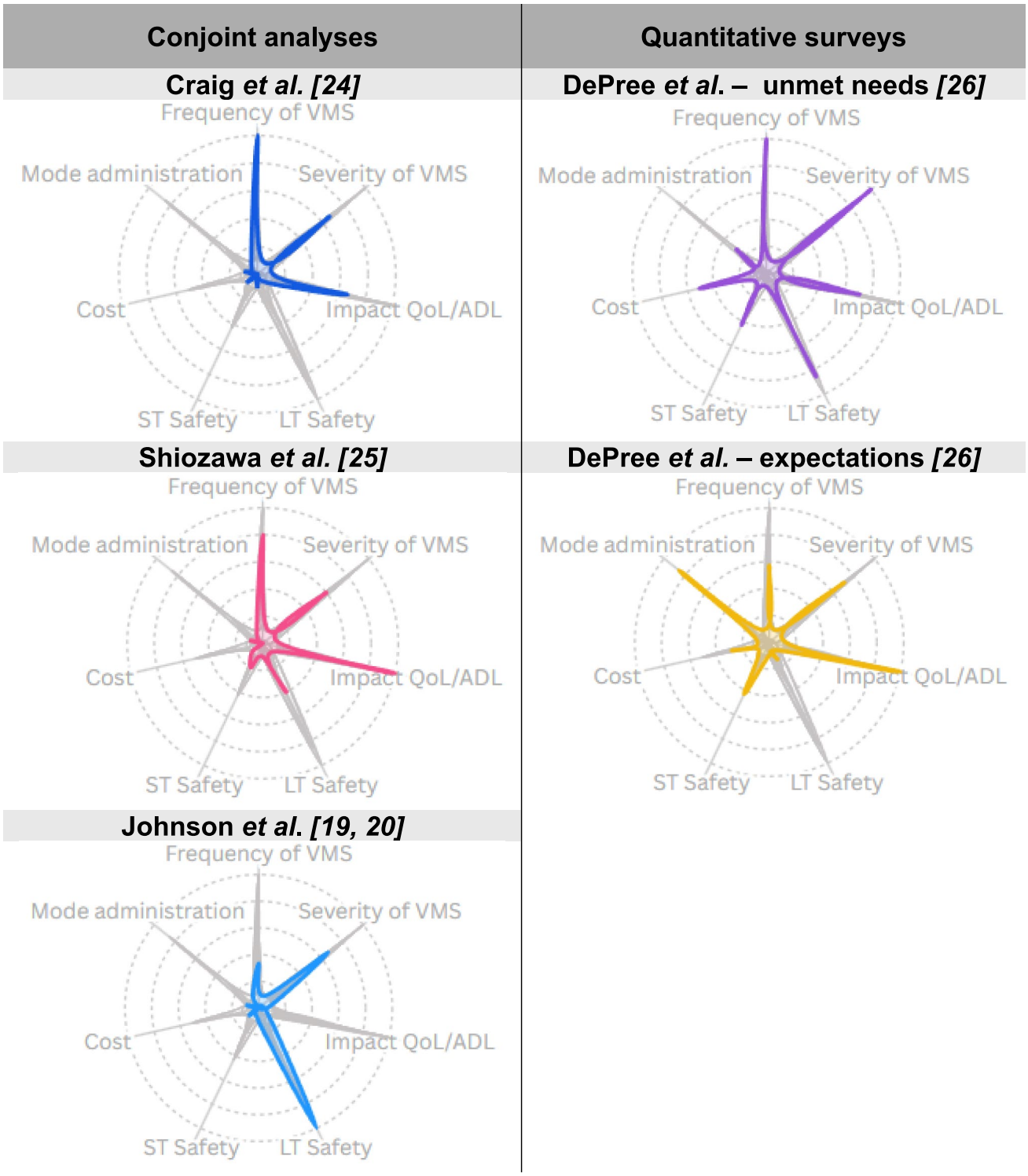
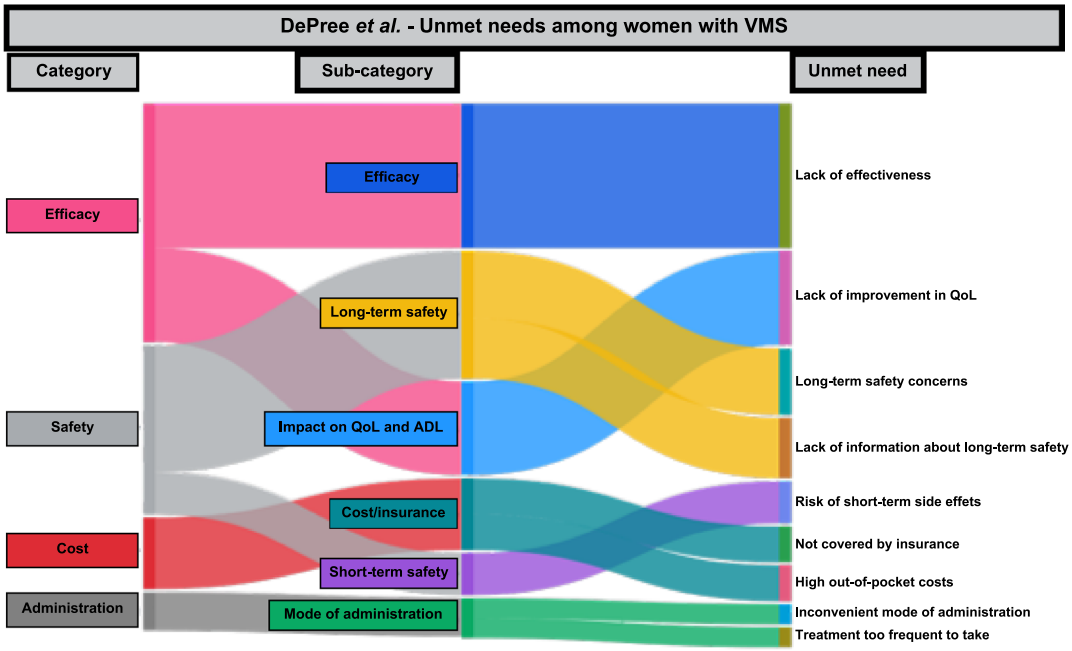
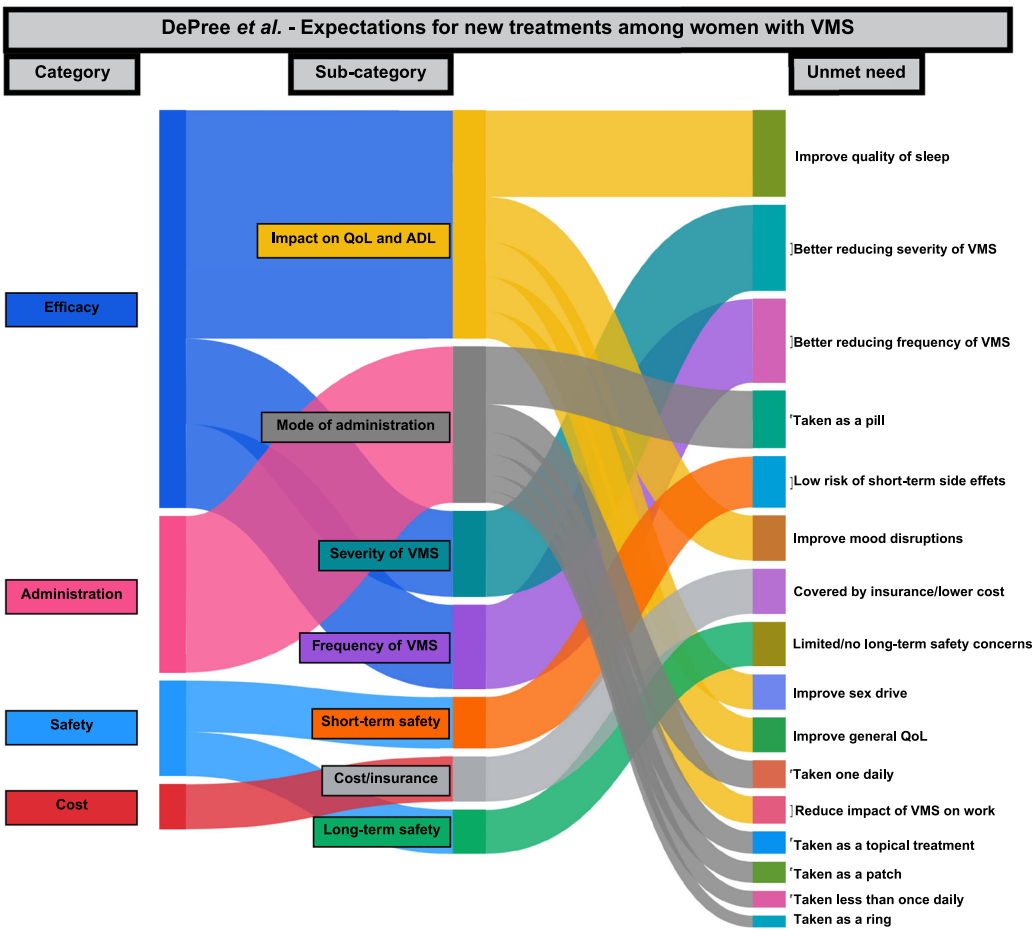


Fig. 4 Preference profiles regarding symptom and treatment preferences. An internal scoring scale was established for each article, with a maximum of 7 points depending on the number of attribute categories included (see attribute definitions). For example, Craig et al., addressing only efficacy attributes, had a 3-point scale; Shiozawa et al., covering five categories, had a 5-point scale. The most important attribute received the highest score, the second received one point less, and so on. Scores were then normalized as percentages. This representation is purely visual and does not display preference weights. Instead, it summarizes which attributes were identified as the most important in each study, on the basis of an internal ranking system

a


DePree et al. - Unmet needs among women with VMS
Category
Sub-category
Unmet need
Efficacy
Safety
Cost
Administration
Efficacy
Long-term safety
Impact on QoL and ADL
Cost/insurance
Short-term safety
Mode of administration
Lack of effectiveness
Lack of improvement in QoL
Long-term safety concerns
Lack of information about long-term safety
Risk of short-term side effets
Not covered by insurance
High out-of-pocket costs
Inconvenient mode of administration
Treatment too frequent to take

b


DePree et al. - Expectations for new treatments among women with VMS
Category
Sub-category
Unmet need
Efficacy
Administration
Safety
Cost
Impact on QoL and ADL
Mode of administration
Severity of VMS
Frequency of VMS
Short-term safety
Cost/insurance
Long-term safety
Improve quality of sleep
Better reducing severity of VMS
Better reducing frequency of VMS
Taken as a pill
Low risk of short-term side effets
Improve mood disruptions
Covered by insurance/lower cost
Limited/no long-term safety concerns
Improve sex drive
Improve general QoL
Taken one daily
Reduce impact of VMS on work
Taken as a topical treatment
Taken as a patch
Taken less than once daily
Taken as a ring

◂**Fig. 5** Quantitative survey categories reported in DePree et al. To enable comparison between conjoint analysis attribute categories and items reported from quantitative surveys, quantitative survey items were classified into categories aligned with those used in the analysis of conjoint analyses. Items were first grouped into categories and then into subcategories. Segment width is proportional to the percentage of reporting: the more frequently an item was reported, the wider its segment. The width of each subcategory corresponds to the sum of its included items; the same applies to categories comprising multiple subcategories. Segments are ordered in decreasing frequency, from the most to the least reported

studies included consistently highlighted the efficacy and long-term safety attributes and items as the most important to women (Fig. 3c).

Vasomotor symptoms represent one of the most prevalent complaints during the menopausal transition, reported by up to 80% of midlife women [5]. While menopause itself is generally associated with reduced QoL scores, VMS significantly contribute to this impaired QoL [27]. Although severity and frequency are central in understanding the impact of VMS, factors such as negative affect, symptom sensitivity, duration, sleep quality, age, and ethnicity also contribute to the overall symptom burden [28]. Indeed, our systematic review revealed that women prioritized the reduction of frequency and severity of VMS, but also treatments that improve sleep quality. This preference underscores the correlation between VMS, QoL, and declined sleep quality [29–31]. Notably, the SMART-2 study revealed a significant linear association between frequency and severity of hot flushes and overall MENQOL scores, all individual MENQOL domains, as well as sleep parameters [30]. In addition, Johnson et al. highlighted that women were willing to make trade-offs between potential risks within the range reported by the WHI and improvements in clinical health states [19]. Authors further demonstrated that women were even willing to sacrifice years of life to avoid intense menopausal symptoms and preferred shorter durations of severe symptoms over prolonged periods of milder ones [20]. These results emphasize the dual burden of VMS severity and duration, as well as the need for a comprehensive overview of VMS and its impact. In addition, these results indicate that women would still trade years of life for symptom relief, even if the WHI may have overestimated the risks. This highlights both the heavy burden of VMS and the incomplete or misleading information from the WHI study.

Experiencing VMS also led to reduced work productivity. One included study reported Work Productivity and Activity Impairment (WPAI) results, showing that women rarely missed work owing to VMS but experienced greater impairment while working [22]. Work and activity impairments worsened with VMS severity. These results are consistent with previous findings indicating that an increase in VMS severity is associated with a decline in productivity at work and at home, as well as an increased reliance on healthcare resources, thereby highlighting the economic burden of menopause [29]. Indeed, menopause has a substantial economic impact on women and society, encompassing not only direct medical costs but also indirect costs such as lost productivity, presenteeism, absenteeism, being downgraded to lower roles, and even quitting or retiring early [32]. A recent study of 4440 working women in the USA found that the yearly cost of lost workdays due to menopause symptoms is about $1.8 billion. When including direct medical expenses, this amount increases to $26 billion. Notably, these figures do not include expenses related to reduced working hours, unemployment, and early retirement [32]. These findings emphasize the importance of identifying strategies that are both clinically and cost-effective for managing VMS.

Safety concerns, in particular long-term safety criteria, were, along with efficacy concerns, the most important to women. Only two conjoint analyses investigated long and short-term safety concerns [19, 20, 25]. Shiozawa et al. [25] specified both short (e.g., weight gain, headache, and nausea) and long-term safety concerns, whereas Johnson et al. [19, 20] considered only long-term risks. Among long-term safety outcomes, only cardiovascular events, cancer, and fractures were represented. However, MHT affects multiple risk outcomes, including cancers (e.g., breast, endometrial, and ovarian cancers), cardiovascular diseases such as venous thromboembolism, musculoskeletal outcomes (e.g., muscle mass and strength, osteoporosis), and cognition and dementia [33]. In fact, conjoint analyses should capture a broader range of safety attributes. In addition, consistent with Johnson et al., DePree et al. found that safety was one of the most important item categories [26]. Notably, quantitative survey approaches, in which no hypothetical choice set was imposed, systematically highlighted both efficacy and safety items as the most important outcomes (e.g., unmet needs and expectations among patients and physicians). This pattern suggests that safety is an important concern for women and indicates the need to include both short- and long-term safety attributes in future conjoint analyses.

Regarding patient preferences for LET, Mattsson et al. [21] identified mode of administration as the most important attribute, and Minkin et al. [23] identified it as the most frequently mentioned item (Fig. 6). Minkin et al. reported the mode of administration, the safety, and the efficacy concerns as the three most reported items, respectively (Fig. 6) [23]. Yet, Mattsson et al. only focused on the mode of administration, smudges/leakages, time of appliance, and cost [21]. Again, when no specific choices are imposed through the conjoint analysis method, women spontaneously report efficacy and safety concerns.

Two quantitative surveys, DePree et al. and Stute et al., also investigated physicians' perceptions and expectations regarding VMS treatments. In both surveys, physicians

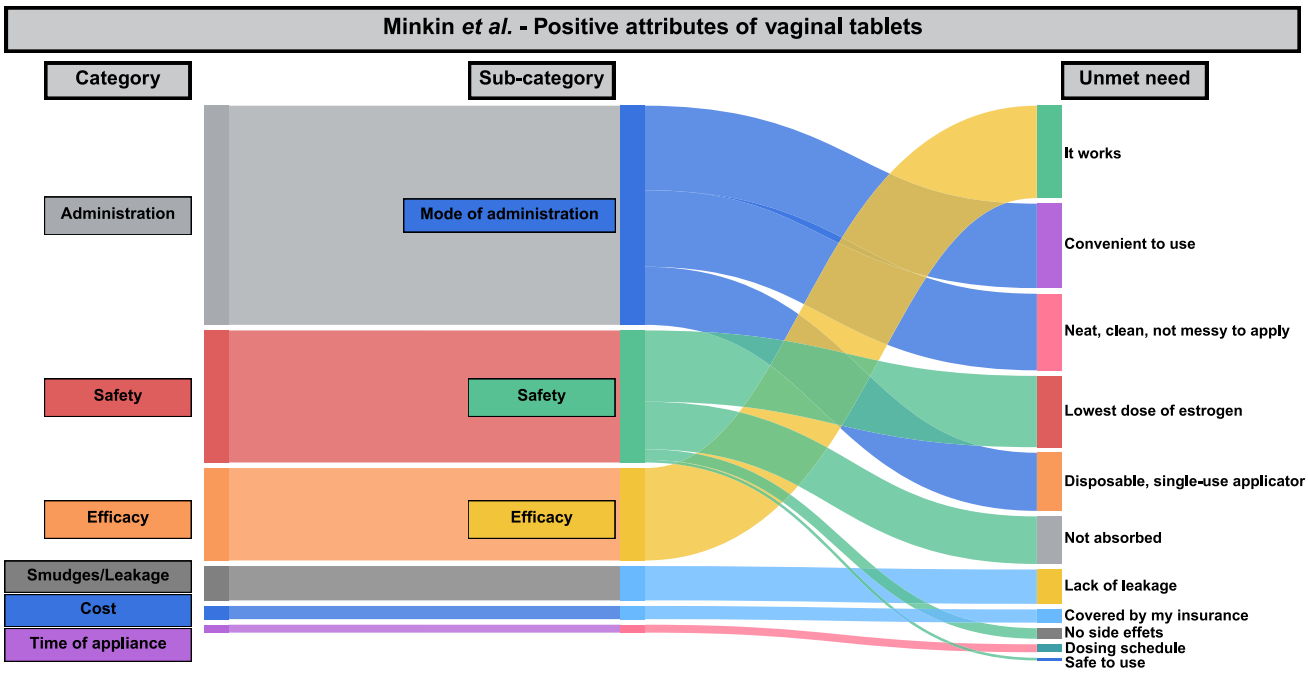


Fig. 6 Positive attributes of vaginal tablets reported by Minkin et al. To enable comparison between conjoint analysis attribute categories and items reported from quantitative surveys, quantitative survey items were classified into categories aligned with those used in the analysis of conjoint analyses. Items were first grouped into categories and then into subcategories. Segment width is proportional to the percentage of reporting: the more frequently an item was reported, the wider its segment. The width of each subcategory corresponds to the sum of its included items; the same applies to categories comprising multiple subcategories. Segments are ordered in decreasing frequency, from the most to the least reported

placed a high value on long-term safety items. Indeed, the primary unmet need of physicians regarding current treatments was long-term safety concerns and the second expectation of physicians regarding new treatments was no long-term safety concerns [26]. Furthermore, the main reason for not prescribing hormonal therapy was patient concerns about long-term risks, while the main reason for not prescribing nonhormonal treatments (e.g., selective serotonin reuptake inhibitors [SSRIs] and serotonin and norepinephrine reuptake inhibitors [SNRIs]) was concerns about the treatment's efficacy [22]. These findings showed that clear, reassuring information about the safety of current treatments remains lacking and affects both patients and healthcare professionals. For instance, an international online survey revealed that only about half of the women felt able to make an informed decision about MHT [34]. In the Vaginal Health: Insights, Views and Attitudes (VIVA) study, half (46%) of all women lacked knowledge about LET [35]. This uncertainty is partly due to the lasting impact of the 2002 WHI studies [36]. In fact, the changes observed in MHT use patterns and the reasons for initiating or discontinuing MHT before and after the WHI, as reported in the SWAN study, demonstrate the rapid widespread dissemination of the WHI's key findings largely due to significant media coverage. However, while these effects have mostly faded, a distorted view of the risks of hormone therapy has persisted, often overshadowing its proven benefits [6]. This has resulted in a decline in MHT use, particularly among symptomatic women in their early menopause, when the VMS are most prevalent, and the risks of MHT are lower [37–39]. Importantly, the US Food and Drug Administration has recently removed the black-box warning from MHT products, signaling a shift toward more evidence-based communication of risks and benefits [8]. This regulatory evolution underscores the need to better understand how clinicians perceive and apply these treatments in practice. Exploring physicians' preferences, therefore, offers valuable insights into how clinical recommendations are interpreted and implemented, and allows for comparison with patients' preferences, ultimately supporting clearer communication and more effective shared decision-making. Furthermore, since current preferences remain heavily influenced by the misinformation from WHI, women's and physicians' perceptions can distort how they value treatment attributes. This underscores the importance of revisiting preference studies in the coming months or years, once perception of the risk has changed.

Both physicians and patients reported low satisfaction with current treatments [22, 25, 26]. In particular, DePree et al. reported that effectiveness is an important unmet need and an expectation for future treatments [26]. MHT is still

recommended as the gold standard for bothersome VMS and urogenital symptoms when no contraindications are present [33, 40]. Other options are available, such as non-hormonal treatments including SSRIs, SNRIs, gabapentin, and clonidine, which are used off label, neurokinin receptor antagonists (e.g., fezolinetant and elinzanetant), and over-the-counter treatments. These treatments are no more effective than MHT [39] and raise questions about their interaction and safety profile, such as SSRI-related adverse events, drug–drug interactions, and the emerging hepatic and neoplasia risks reported with fezolinetant. [41, 42]. MHT not only supports bone health and significantly reduces the risk of osteoporosis-related fractures but also offers cardiovascular protection when initiated within 10 years of the final menstrual period [43]. Moreover, after re-evaluation following the WHI study results, MHT for VMS remains cost-effective compared with no therapy [44]. Nonetheless, the potential economic advantage of MHT, alongside its clinical benefits, must be considered in light of individual risk profiles and patient preferences [6].

An important aspect to mention is that none of the included conjoint analyses accounted for respondents' health or numeric literacy, potentially biasing preference estimates [45, 46]. Moreover, the methodology used to communicate risk to women varied across conjoint analyses. Among the two studies that incorporated safety as an attribute, Shiozawa et al. reported only absolute values, whereas Johnson et al. reported both absolute and relative values. Johnson et al. reported that women who received the absolute risk version of the survey were willing to accept greater heart attack risks than those who received the relative risk version. Previous studies have suggested that discussing figures in absolute risk may help patients better evaluate the potential risks and benefits [47, 48]. However, the original publication of the WHI, as well as much of the publicity that followed, presented the results in the form of relative risk, which may have led patients to misestimate the risks compared with the benefits [49]. In addition, one way to facilitate comprehension is to visually separate baseline from treatment risk. However, none of the included conjoint analyses appeared to use graphical risk displays (e.g., pictograms) to support understanding of quantitative risk [48].

This review is limited by the small number of conjoint analyses, especially the few DCEs ($n = 2$), and the limited number of quantitative surveys included, particularly for evaluating preferences toward LETs. In addition, two of the eight included studies were identified through manual searching rather than database queries, which may highlight a potential limitation in the search strategy's sensitivity. However, manual searching remains a crucial and integral part of comprehensive evidence identification process. Nevertheless, including both conjoint analyses and quantitative surveys allows a closer approximation of real-world preferences while revealing each method's limitations relative to the other. Although conjoint analyses and quantitative surveys varied in methodology, transparent and high-quality reporting across studies enabled the harmonization of key attributes and items and the development of comparable metrics, including alluvial diagrams and radar charts.

5 Conclusions

This systematic review highlights substantial heterogeneity in methods used to assess women's preferences for the management of menopausal symptoms, which hampers the ability to synthesize findings. Significant gaps still exist in understanding treatment preferences for menopausal care. These gaps could be filled through health technology assessment approaches, such as robust conjoint analyses. Although DCEs are the most common method for eliciting health care preferences, they are rarely used in this area. Future conjoint analyses, especially DCEs, should include complete attribute sets that cover the full range of menopausal care, including efficacy, safety outcomes, and current information on treatment risks and benefits to prevent misinformation. Simultaneous evaluation of both physician and patient preferences may help align clinical advice, patient education, and women's values. In addition, since most existing studies are based in the USA, there is a need for more conjoint analyses in Europe, Asia, Africa, Oceania, and Latin America.

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Data availability Data are available under reasonable request to the corresponding author.

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